

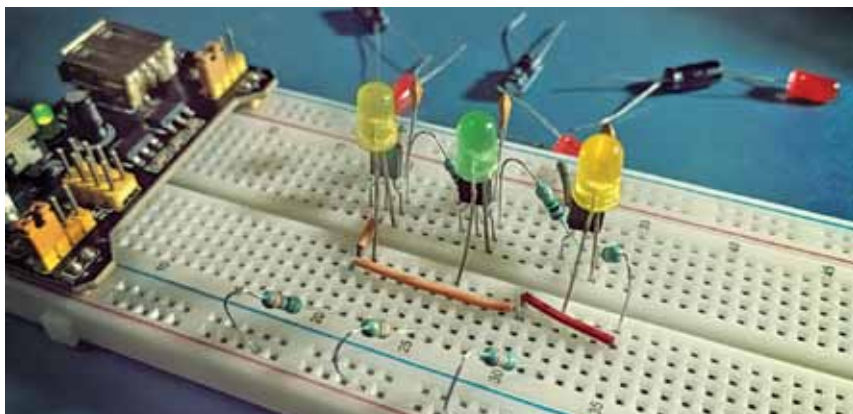


Fig. 3: Author's prototype

Construction and testing

The construction of the device can be divided into several stages. To begin with, it is recommended to test the circuit on a breadboard. Connect the USB cable's +5V and GND wires to the power rails of the breadboard, and then place the MOSFETs, resistors, capacitors, and LEDs according to the circuit diagram in Fig. 2. This allows easy troubleshooting and

experimentation with different capacitor values to fine-tune the blinking speed. The author's prototype is shown in Fig. 3.

After the breadboard prototype operates successfully, the circuit can be transferred to a perfboard or PCB. While soldering, leave sufficient spacing for the LEDs so that they can be mounted inside the vase. Thin insulated wires assist in positioning them neatly.

Choose a transparent or frosted

vase for a better glow. Secure the LEDs inside using adhesive or clips, ensuring they are evenly spaced. Place the main circuit board at the bottom or in a small case to keep it safe and hidden. The circuit operates on 5V USB, consuming minimal current and generating no heat, making it safe for continuous use.

To test the device, plug it into a laptop, charger, or power bank. The LEDs should blink in sequence, creating a rotating effect. If this does not occur, check the wiring, MOSFET pin connections, and RC section values. Capacitor values can be varied to adjust the blink speed. After testing, secure the board and LEDs neatly inside the vase for the best visual effect.

Once assembled, the device serves as both a decorative lighting element and a practical demonstration of ring oscillator operation. **EFY**



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